

Elevation of extracellular adenosine mobilizes haematopoietic progenitor cells and granulocytes into peripheral blood and enhances the mobilizing effects of granulocyte colony-stimulating factor

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Abstract: We tested the capabilities of drugs elevating extracellular adenosine and of granulocyte colony-stimulating factor (G-CSF), given alone or in combination, to mobilize haematopoietic progenitor cells for granulocytes and macrophages (GM-CFC) and granulocytes into peripheral blood. Elevation of extracellular adenosine was induced by joint administration of dipyridamole (DP), a drug inhibiting the cellular uptake of adenosine, and adenosine monophosphate (AMP) serving as an adenosine prodrug. DP + AMP, G-CSF or all these drugs in combination were administered either singly or repeatedly in a 4-d treatment regimen. Elevation of extracellular adenosine was found to mobilize significantly both GM-CFC and granulocytes after both single and repeated administration of DP + AMP. These results show that the elevation of extracellular adenosine presents a potent mechanism for mobilization of GM-CFC and granulocytes into the blood. When the combination of DP + AMP + G-CSF was given under the 4-d regimen, the mobilizing effects of its administration were additive when compared with those of DP + AMP alone or G-CSF alone. The observed ability of the drugs elevating extracellular adenosine to enhance the mobilizing action of G-CSF points out possible practical utilization of the findings presented here. This conclusion is further supported by the results of an additional experiment which indicate that blocking of haemodynamic side effects of drugs elevating extracellular adenosine by noradrenaline does not suppress their mobilizing effects.

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In the present, the transplantation of peripheral blood progenitor cells (PBPCs) is widely used in protection against haematologic toxicity of anti-cancer chemotherapy. More rapid recovery of granulocytes and platelets is achieved with PBPCs in comparison with bone marrow transplantation (1). Granulocyte colony-stimulating factor (G-CSF) has been shown to be an effective mobilization factor in the mice (e.g. 2), as well as in the man (e.g. 3). Numerous experimental settings comprising combined administration of G-CSF with various

cytokines or growth factors, e.g. stem cell factor (4, 5), interleukin-3 (6), FLT-3 ligand (7, 8), or megakaryocyte growth and development factor (9) have been successfully tested. Nevertheless, in current clinical practice, G-CSF is used most commonly as the only mobilization-stimulating agent; only sometimes it is combined with cyclophosphamide (10).

Much effort has been devoted to the elucidation of the mechanisms of the PBPCs-mobilizing effects of G-CSF and other substances which has brought

important findings concerning mobilization of haematopoietic progenitor cells into peripheral blood in general. Thus, e.g. Papayannopoulou and Nakamoto (11) have emphasized the role of adhesion molecules, namely VLA-4. On the basis of the results of studies on mice deficient in the G-CSF, Link (12) concluded that the receptor for G-CSF is required for the mobilization not only in response to G-CSF but also in response to other agents, namely cyclophosphamide or interleukin-8. Lapidot and Petit (13) have stressed the importance of chemokines, such as stromal derived factor-1, and their receptors. In studies on rats, Iversen *et al.* (14) have shown that administration of G-CSF caused not only the stimulation of granulocyte formation but also a significant increase in the bone marrow blood flow; this finding suggests that processes of proliferation and mobilization of haematopoietic progenitor cells might be modified also by vascular changes. Taken together, it follows from these and other studies that mobilization of haematopoietic progenitor cells into the peripheral blood is a complex multistep process which may be influenced by various interventions.

Experimentally induced elevation of extracellular adenosine, achieved by combined administration of dipyridamole (DP), a drug preventing the cellular uptake of adenosine and preserving it in the extracellular space (15), and adenosine monophosphate (AMP), serving as an adenosine prodrug, has been demonstrated in numerous studies carried out in our laboratories to control positively haematopoiesis in mice. Thus, pre-irradiation administration of DP + AMP has been found to act radioprotectively and to stimulate haematopoiesis in mice exposed to single (16) or repeated (17, 18) doses of ionizing radiation. Moreover, we have observed that the elevation of extracellular adenosine potentiates the action of G-CSF. The joint administration of the drugs elevating extracellular adenosine and of G-CSF enhanced haematopoiesis in normal (19) or gamma-irradiated mice (20, 21), as well as in mice with myelosuppression acquired either by treatment with a cytostatic drug, 5-fluorouracil, alone (22, 23), or by combined treatment with another cytostatic drug, carboplatin, and ionizing radiation (24). In a study aimed at clarifying the mechanism of the adenosine action on haematopoiesis, we have demonstrated that administration of DP + AMP to mice increases the cycling of haematopoietic progenitor cells in the bone marrow (25).

Based on the above given findings we have decided to investigate the ability of extracellular adenosine to mobilize haematopoietic progenitor cells and granulocytes into blood. Such a study could lead to better understanding of the regulatory

role of adenosine in haematopoiesis and, at the same time, to contribute to the efficacy of procedures which utilize the mobilization phenomenon for therapeutic purposes. For experimental elevation of extracellular adenosine in mice, we used joint administration of DP + AMP which was described previously and proved to be efficient (16–25). Mobilization effect was recorded by the determination of granulocyte–macrophage colony-forming cells (GM-CFC) and granulocytes in the peripheral blood. Effects obtained by the use of DP + AMP were compared with those induced by G-CSF. In addition, the effects of the combined administration of DP + AMP with G-CSF were investigated.

Material and methods

Mice

Male mice (CBAx57BL/10F₁) from the breeding facility of Faculty of Medicine, Masaryk University (Brno, Czech Republic), 3-months old and weighing in average 30 g were used. A standardized pelleted diet and HCl-treated tap water (pH 2–3) were given *ad libitum*. The mice were kept under controlled conditions (light : dark, 12 : 12) and the temperature was maintained at 22 ± 1°C. Animal experiments were conducted according to the principles given in the 'Guide for the Care and Use of Laboratory Animals' issued by the Czech Society for Laboratory Animal Science.

Drugs and study design

Dipyridamole (DP; Sigma, St Louis, MO, USA) was dissolved in 0.4% tartaric acid and injected subcutaneously (s.c.) at a dose of 2 mg per mouse (66 mg/kg b.w.) in a volume of 0.4 mL. Adenosine 5'-monophosphate from yeast (AMP; Sigma) was diluted with saline and administered intraperitoneally (i.p.) at a dose of 5 mg per mouse (165 mg/kg b.w.) in a volume of 0.2 mL. Recombinant human G-CSF (Neupogen, F. Hoffman La Roche Ltd, Basel, Switzerland) was dissolved in 5% glucose and injected s.c. at a dose of 3 µg per mouse (99 µg/kg b.w.) in a volume of 0.1 mL. In the combination treatment, DP was administered 20 min before AMP, G-CSF was administered 30 min after AMP. The corresponding drug vehicles were used for control injections. The doses of the drugs were based on our earlier experience (16–25). It has to be mentioned that the relatively high dose of DP used is substantiated by the poor sensitivity of mice to this nucleoside transport inhibitor (26). This dose presents in mice about 1/40 of the LD₅₀ (27).

For evaluating the ability of the drugs to mobilize haematopoietic progenitor cells into the peripheral blood, the effects of DP + AMP, G-CSF or the DP + AMP + G-CSF combination were tested in three treatment regimens: (i) all the drugs were given in single doses, peripheral blood sampling was performed 6 h after the administration of G-CSF; (ii) all the drugs were given once daily for 4 d, sampling was performed 6 h after the last injection of G-CSF; (iii) all the drugs were given once daily for 4 d, sampling was performed 24 h after the last injection of G-CSF.

For evaluating the role of the action of DP + AMP on the vascular system of the bone marrow in the mobilization effects observed, an additional experiment was performed in which DP + AMP were given as mentioned above in single doses either without or with noradrenaline (Léčiva, Praha, Czech Republic) in a dose of 0.042 mg per mouse (1.4 mg/kg b.w.) in a volume of 0.2 mL administered s.c. immediately before AMP. Such a dose of noradrenaline has been found in our earlier experiments (28) to eliminate the cardiovascular effect of DP + AMP in mice. Sampling was performed 6 h after the administration of AMP.

Haematological methods

Mice were anaesthetized i.p. with a Narcamon/Rometar solution in the ratio of 2.63 : 1 (5% Narcamon and 2% Rometar, Spofa, Prague, Czech Republic) and peripheral blood was obtained by cardiac puncture. For assessment of numbers of GM-CFC per millilitre of the peripheral blood, GM-CFC were assayed using a semisolid 0.33% agar technique (29, 30). Briefly, Iscove's Modified Dulbecco's Medium (IMDM; Sigma) was supplemented with 20% fetal calf serum, 5% conditioned medium from mouse interleukin-3-producing myeloma cell line (Institute of Hematology, Charles University, Prague, Czech Republic) and 0.33% boiled agar Noble (Difco Laboratories, Detroit, MI, USA). Heparinized (500 U/mL) peripheral blood of each single mouse was plated in triplicate in a volume of 0.1 mL/1.0 mL IMDM and incubated at 37°C in a humidified atmosphere containing 5% CO₂. GM-CFC colonies were scored after 7-d incubation as colonies containing 50 or more cells following administration of a few drops of 20% acetic acid on the surface of agar for dissolving erythrocytes. For evaluation of numbers of granulocytes per microlitre of the peripheral blood, numbers of leukocytes were determined using a Coulter Counter (model ZF, Coulter Electronics, Luton, UK) and differential counts were performed on smear

preparations stained with the May-Grünwald-Giemsa method.

Statistics

The data are given as geometric mean $\pm$ SEM. The results were subjected to Kruskal-Wallis ANOVA and for comparison of two-group data the Mann-Whitney *U*-test was used. To correct for multiple comparisons, the Holm (31) correction was applied. The significance level was set at $P < 0.05$.

Results

Two main questions have been raised in the evaluation of the results. They comprised the investigations of the effects of drugs elevating extracellular adenosine (DP + AMP) on the mobilization of GM-CFC and granulocytes into blood, and secondly, the verification of their possible use in the enhancement of the mobilizing effect of G-CSF. Based on this experimental intention, four statistical comparisons have been used in describing the results, namely groups DP + AMP alone vs. controls, G-CSF alone vs. controls, DP + AMP + G-CSF vs. DP + AMP alone, and DP + AMP + G-CSF vs. G-CSF alone.

Effects of the drugs on GM-CFC mobilization

Figure 1 shows the mean numbers of GM-CFC per millilitre of the peripheral blood in the experimental regimens designated as A, B, and C. When the drugs were given only once (regimen A), both DP + AMP alone and G-CSF alone induced significant mobilizing effects attaining 359% and 183% of controls, respectively. The combination of DP + AMP + G-CSF retained the high mobilization potential (284% of controls). However, this effect was significantly lower compared with the effect of DP + AMP alone. This underadditive effect can be due to some antagonistic action of the components of the drug combination and remains to be further analysed. Experimental regimens B and C, using the 4-d treatment, have shown significant mobilizing effects for both DP + AMP alone (147% and 226% of controls in regimens B and C, respectively) and G-CSF alone (258% and 307% of controls in regimen B and C, respectively). The combination of DP + AMP + G-CSF induced the highest mobilization effects (342% and 523% of controls in regimens B and C, respectively). These effects were significantly different when compared with those obtained after DP + AMP alone and G-CSF alone. It can be calculated that under repeated administration,

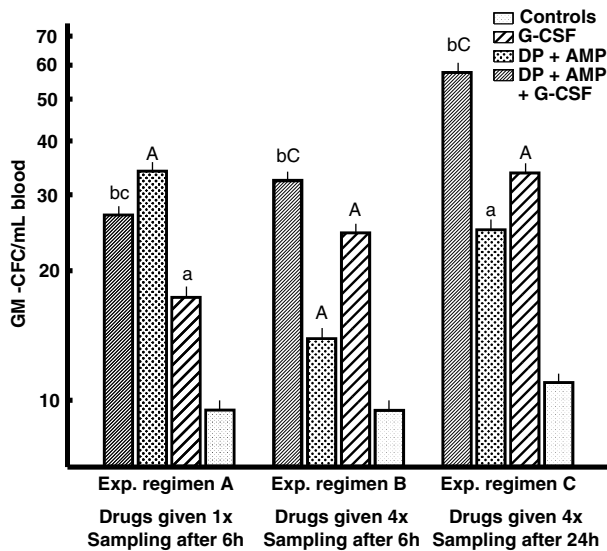


Fig. 1. Numbers of GM-CFC per millilitre of the peripheral blood in differently treated groups of mice. Data are given as mean $\pm$ SEM for 10 mice per group. Statistical significance: A and a, $P < 0.01$ and $P < 0.05$, respectively, in comparison with controls. B and b, $P < 0.01$ and $P < 0.05$, respectively, in comparison with G-CSF-treated mice. C and c, $P < 0.01$ and $P < 0.05$, respectively, in comparison with mice treated with DP + AMP.

DP + AMP and G-CSF act on the mobilization of GM-CFC at least additively.

Effects of the drugs on granulocyte mobilization

Figure 2 depicts mean numbers of granulocytes per microlitre of peripheral blood observed in the same mice as used in the experiment shown in Fig. 1. Experimental regimen A, using single administration of drugs, did not indicate significant influence of G-CSF alone, but demonstrated high and significant mobilization effect of DP + AMP alone (269% of controls). This effect did not differ from that obtained after combined administration of all drugs (286% of controls). Effects observed in experimental regimens B and C using repeated administration of drugs indicated significant mobilization effects for both DP + AMP alone (212% and 190% of controls in regimens B and C, respectively) and G-CSF alone (160% and 136% of controls in regimens B and C, respectively). The combination of DP + AMP + G-CSF induced the highest effects (320% and 282% of controls in regimens B and C, respectively). These effects were significantly different when compared with those obtained after DP + AMP alone and G-CSF alone. It can be calculated that under repeated administration, DP + AMP and G-CSF act on the mobilization of granulocytes slightly more than additively.

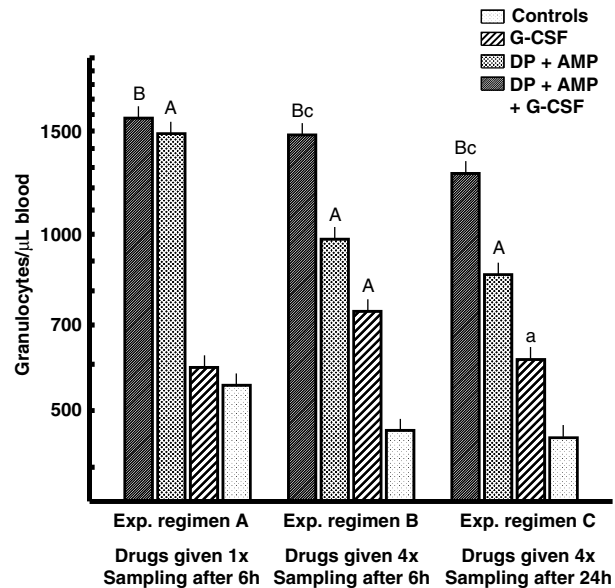


Fig. 2. Numbers of granulocytes per microlitre of the peripheral blood in differently treated groups of mice. Data are given as mean $\pm$ SEM for 10 mice per group. Statistical significance: A and a, $P < 0.01$ and $P < 0.05$, respectively, in comparison with controls. B and b, $P < 0.01$ and $P < 0.05$, respectively, in comparison with G-CSF-treated mice. C and c, $P < 0.01$ and $P < 0.05$, respectively, in comparison with mice treated with DP + AMP.

Effects of noradrenaline on DP + AMP action

In additional experiments, the possible influence of the vascular and haemodynamic effects of extracellular adenosine on its mobilization effects was investigated. Mobilization of granulocytes into blood, induced by DP + AMP given either alone or in combination with noradrenaline at a dose counteracting the vasodilatory and haemodynamic action of adenosine (26), was examined after single administration of the drugs. As shown in Table 1, noradrenaline did not influence the mobilizing effect of DP + AMP. Thus, the role of haemodynamic action of adenosine in these mobilization effects could be ruled out.

Discussion

It follows from the results that elevation of extracellular adenosine by the administration of DP + AMP represents a potent mechanism mobilizing GM-CFC and granulocytes into the blood. As shown, the mobilizing effects are manifested under both single and repeated administration of drugs. However, the mechanisms participating in the mobilization effects can differ in these two experimental regimens. Under single administration of the drugs and the early determination of the mobilization response, the effects can depend only on the potency of drugs to mobilize cells from the

Table 1. Numbers of granulocytes per microlitre of the peripheral blood in mice treated with DP + AMP, noradrenaline (NA), or with the combination of all these drugs

	DP + NA + AMP	DP + AMP	NA	Controls
Granulocytes/ μ L blood	1634 $\pm$ 207*	1544 $\pm$ 69*	739 $\pm$ 134	468 $\pm$ 73

All drugs were administered in a single dose, sampling was performed after 6 h. Data are given as mean $\pm$ SEM for five mice per group. Statistical significance: * $P < 0.05$ in comparison with controls. Differences between the groups DP + NA + AMP and DP + AMP, as well as between the groups NA and controls, are insignificant.

storage pools which are not yet influenced by the activation of the proliferative cell compartments. Another situation occurs under repeated administration of drugs. Here, the mobilization can be attributed in part to an overall increase in the size of the progenitor cell pool because of the proliferative effects of the drugs. We have shown earlier that the same experimental regimen as used here, i.e. 4-d treatment of mice with the same doses of DP + AMP and G-CSF, is effective in enhancing bone marrow granulopoiesis as manifested in GM-CFC and granulocytic cell counts (19). Thus, under repeated administration, not only the rapid mobilization ability, but also the proliferation enhancing activity can contribute to the mobilization response observed after DP + AMP. As also G-CSF activates proliferation in the bone marrow granulopoietic compartments, it is understandable that the repeated administration of DP + AMP with G-CSF induces additive mobilization effects. Especially these results point out the possible practical utilization of the presented findings.

It is difficult to assess the role of different steps participating in the mobilization activity of the repeatedly administered drugs. However, there seem to exist differences in the mobilized GM-CFC and granulocyte counts between mice treated repeatedly with G-CSF alone or DP + AMP alone. It can be inferred that G-CSF favours the release of GM-CFC, while DP + AMP act more effectively on the release of granulocytes. For example, in the regimen B the enhancement of GM-CFC in blood by G-CSF alone (258% of controls) surpassed that induced by DP + AMP alone (147% of controls) by a factor of 1.75. On the contrary, enhancement of granulocytes in blood by DP + AMP alone (212% of controls) surpassed that induced by G-CSF alone (160% of controls) by a factor of 1.32. This might indicate a more effective role of amplifying mechanisms in the progenitor cell pool in case of G-CSF and higher effectiveness of the 'pure' mobilizing mechanisms in case of DP + AMP.

The action of adenosine on cell mobilization remains to be an open problem. It has to be noted here that extracellular adenosine is believed to exert its cellular effects *via* activation of G protein-coupled adenosine receptors (32) and that some of the recognized effects of adenosine receptor signal-

ling can be relevant to the mobilization phenomenon. The ability of adenosine signalling to increase the size of the progenitor cell pool because of the enhanced cell cycling has been already mentioned (19, 25). In addition, it has been shown that activation of A₂ adenosine receptors inhibits the adherence of granulocytes to endothelium (33) and that the production of interleukin-8, a potent and rapid mobilizer of progenitor cells and of granulocytes into blood (34), can be increased by the activation of A_{2b} adenosine receptor subtype (35). Interestingly, a stimulation of A₃ adenosine receptor by their specific agonist has been found to enhance the production of endogenous G-CSF in mice (36). Interleukin-8 and G-CSF as possible effectors of the elevated extracellular adenosine have been shown to induce proteolytic cleavage of adhesion molecules and could thus contribute by these mechanisms to the mobilization of bone marrow cells (37, 38).

The aim of the separate experiment investigating the joint action of DP + AMP with noradrenaline was to explore the possibility that the effects of extracellular adenosine on the mobilization of cells from the bone marrow are mediated *via* its systemic haemodynamic action. It is known that adenosine induces vasodilation in all vascular beds except the kidney through the activation of adenosine A₂ receptors located on both the endothelial cells and the underlying vascular smooth muscles (39). As shown and reviewed by Iversen (40), the marrow blood flow can play an important role in maintaining an appropriate marrow function and its increase mediated by vasodilation occurs under accelerated haematopoiesis. Direct links of the marrow blood flow to haematopoietic functions, including those of cell release, have not yet been firmly established, but cannot be excluded. Our earlier experiments on mice (28), as well as those performed on rabbits (41, 42), have demonstrated that the haemodynamic responses to adenosine administration could be inhibited by the counteracting effects of noradrenaline. Thus, the finding that administration of noradrenaline at a proper dose inhibiting the haemodynamic response to adenosine does not influence the mobilizing effects of DP + AMP makes the role of such an action of adenosine in the mobilization effects unlikely. The importance of these findings is evident also from

the practical point of view. Despite potential clinical applications of adenosine, its use has been limited as a result of the adverse side effects manifested themselves as hypotension. Minimization of these side effects is thus desirable. The evidence that the positive adenosine action on the mobilization of cells into blood is preserved even under suppression of the side effects, brings our findings nearer to clinical applications. However, further studies are required to determine whether PBPCs collected after activation of adenosine receptors can produce efficient long-term engraftment in recipients.

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